

Business Process Automation

with Power Automate and Copilot Studio Agents

WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–4

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W
Version 6.0 (25 Jul 2026) · www.tertiarycourses.com.sg · <https://lms-tms.tertiaryinfotech.com/>

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

DELIVERS

WSQ courses on AI, business automation, data science and software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Power Automate, Copilot Studio or other automation tools (if any).
- One repetitive business process you would love to automate after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

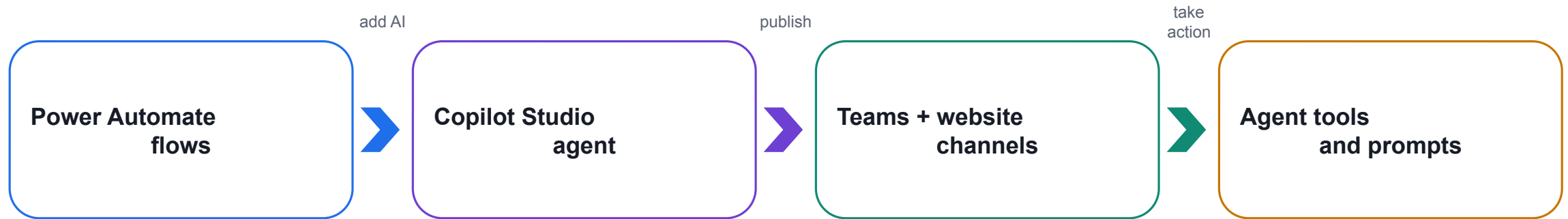
5

Be punctual; return from breaks on time.

6

75% attendance is required.

What You'll Learn



Day 1 — automate

Build reliable trigger → action flows without an AI agent.

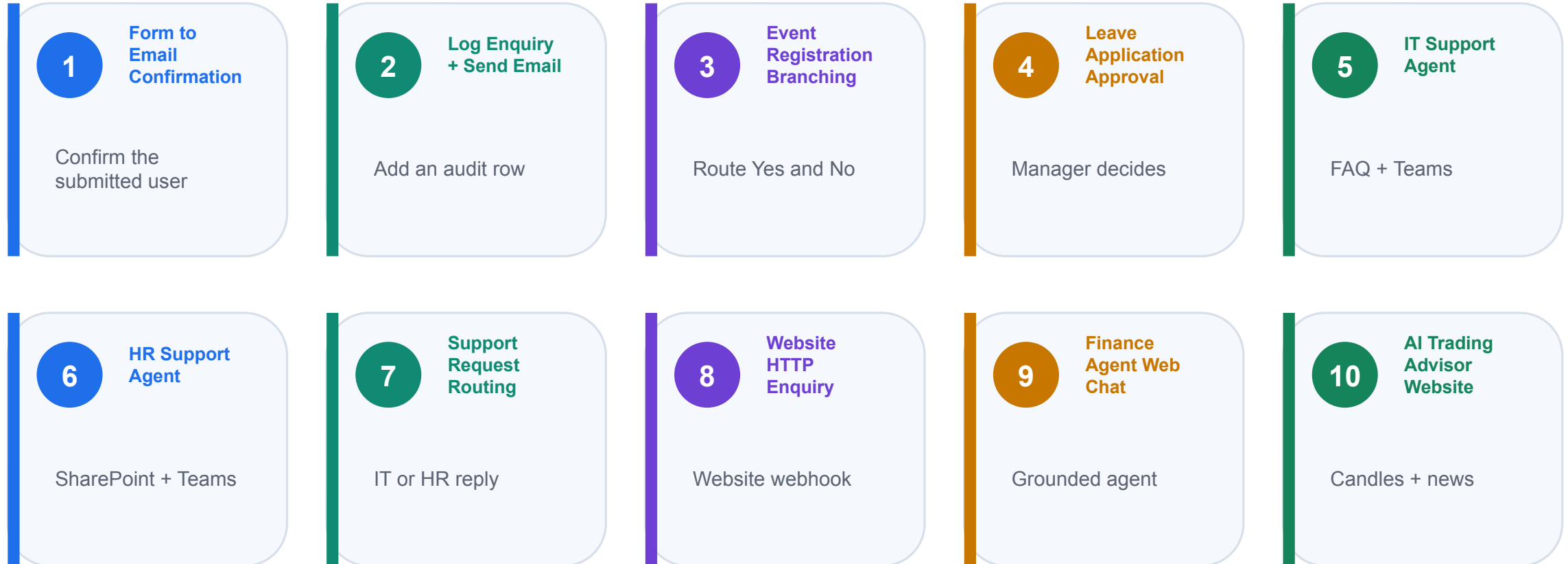
Day 2 — converse

Create a grounded agent that understands and confirms user intent.

Integrate — act

Let the agent call deterministic or prompt-based Power Automate tools.

Your 2-Day Journey



Day 1 builds Forms, flows and support agents. Day 2 adds webhooks and the Finance Advisor capstone.

How This Course Works

- 1 Learn the concept and decision rule before opening the product
- 2 See the concept as a visual business pattern and worked example
- 3 Apply the pattern in the next numbered hands-on lab
- 4 Capture run history, email, workbook, agent or browser evidence
- 5 Reuse each working capability in the next lab instead of starting over

Lesson Plan — 2 Days, 9:00am–6:00pm

D1 Day 1 — Build and route

9:00–10:40 Concepts + flow types
10:55–3:15 Labs 1–4: Forms, Excel, event branch, approval
3:30–5:50 Agent concepts + Labs 5–7
5:50–6:00 Evidence check and recap

D2 Day 2 — Connect and advise

9:00–10:05 HTTP requests and webhooks
10:05–12:15 Lab 8 + Finance Agent foundation
1:15–4:00 Labs 9–10: web chat and trading advisor
4:00–6:00 Written + practical assessment

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment & Funding

1

Written

1 hour · open book
Modules 1–4 and the lab concepts

2

Practical

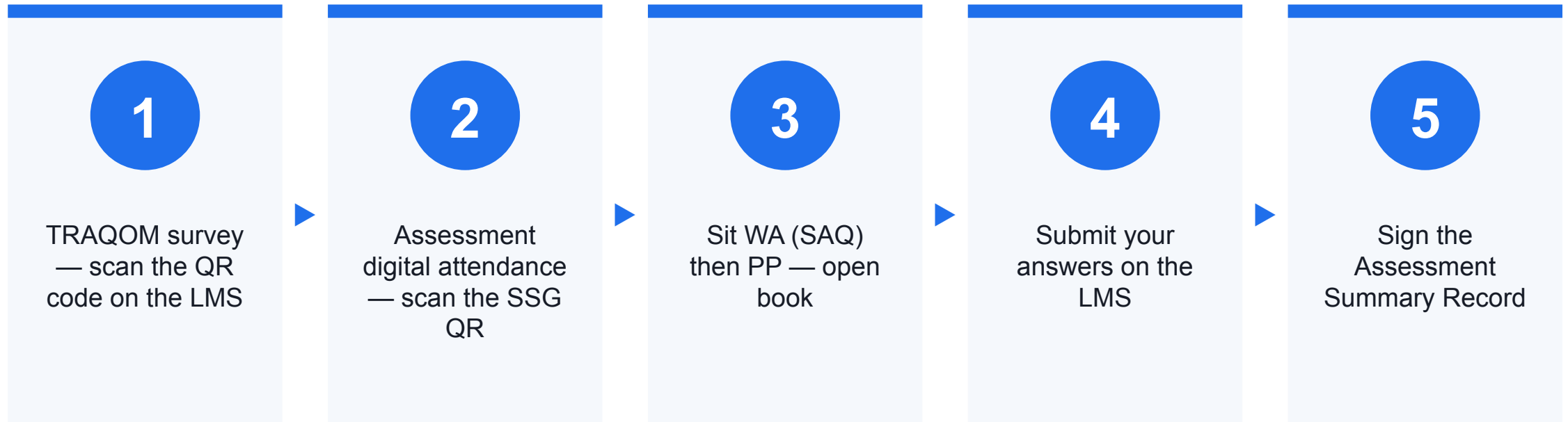
1 hour · open book
Build and verify a working flow and agent

3

Evidence

Submit on the LMS
Competent result + attendance requirement

Assessment Flow



Access the Hands-On Labs

1

10 labs

Lab 0 setup plus Labs 1–10 and four concept modules.

2

Ready assets

Excel tables, searchable PDFs, HTML pages and JSON schemas.

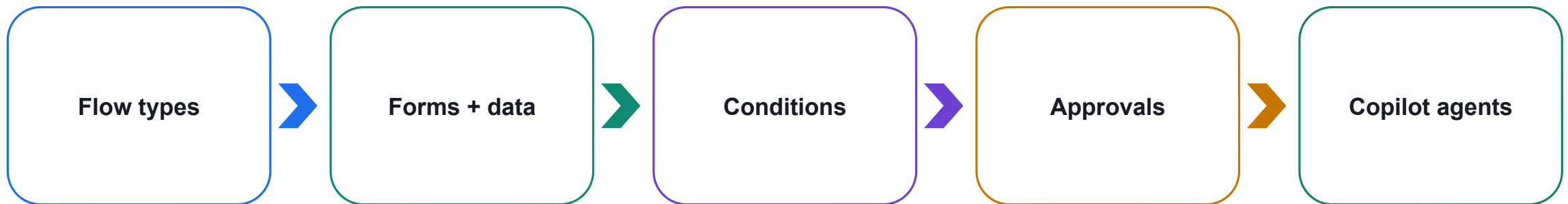
3

Use the guide

Open each index.md or the compiled Learner Guide and work in order.

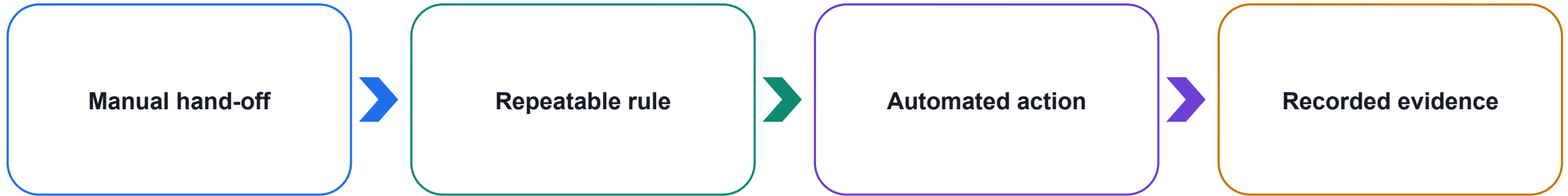
Day 1 — Automate, Decide and Add Agents

DAY 1



Learn each automation concept, apply it in sequence, then connect specialised agents.

What Business Process Automation Changes



Teaching point

Automate stable, repetitive work; keep judgement, exceptions and accountability visible.

Choose Work Worth Automating

1

Repetitive

The same steps occur frequently and consume attention.

2

Rule-based

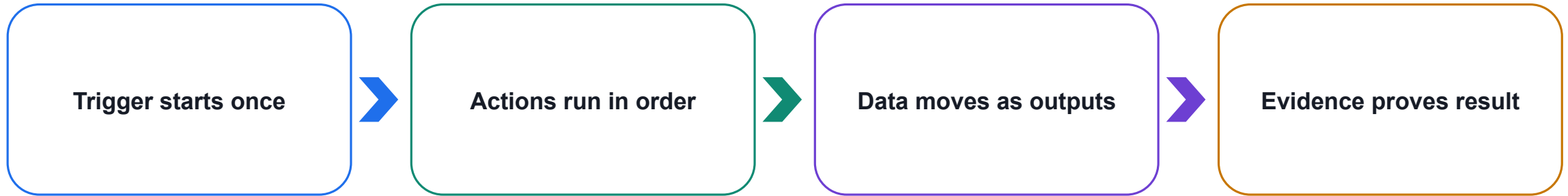
Inputs and outcomes can be expressed as clear business rules.

3

Traceable

The process benefits from timestamps, status and run history.

Every Flow Uses Trigger → Actions → Output



Teaching point

A flow must have exactly one trigger and at least one action before it can save successfully.

Three Cloud Flow Types

1

Instant flow

A person deliberately starts it from a button, app or manual run.

2

Scheduled flow

A Recurrence trigger starts it at a defined time or interval.

3

Automated flow

A connected-service event starts it when new information arrives.

Instant Cloud Flow

1

Start

A user selects Run, presses a button or invokes the flow from an app.

2

Use when

The exact start time depends on a person's decision.

3

Example

A staff member submits a one-off request or launches a controlled test.

Scheduled Cloud Flow

1 Start

A Recurrence trigger reaches a date, time or interval.

2 Use when

Work must happen even when no new business event arrives.

3 Example

Send a weekday reminder at 9:00 AM Singapore time.

Automated Cloud Flow

1 Start

A form, email, file, record or message event occurs.

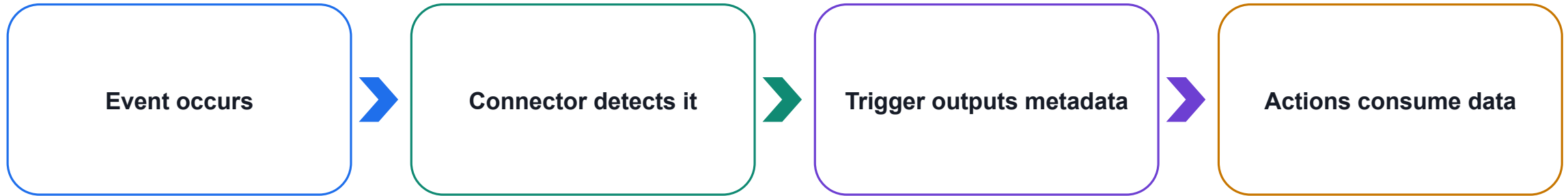
2 Use when

The process must react immediately to new information.

3 Example

A Microsoft Forms response starts Labs 1–4 and 7.

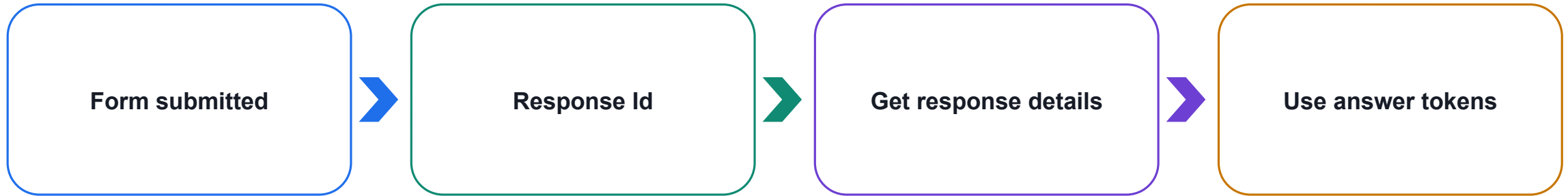
A Trigger Is a Contract with the Starting System



Teaching point

Choose the trigger by asking who or what owns the start event: person, clock or system.

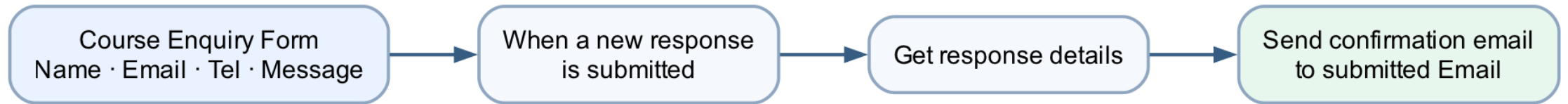
The Microsoft Forms Trigger Pattern



Teaching point

The trigger identifies a response; Get response details retrieves the actual answers.

Lab 1 — Form to Email Confirmation



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 1 — Design the Form

1

Identity

Name and Email identify the requester and recipient.

2

Contact

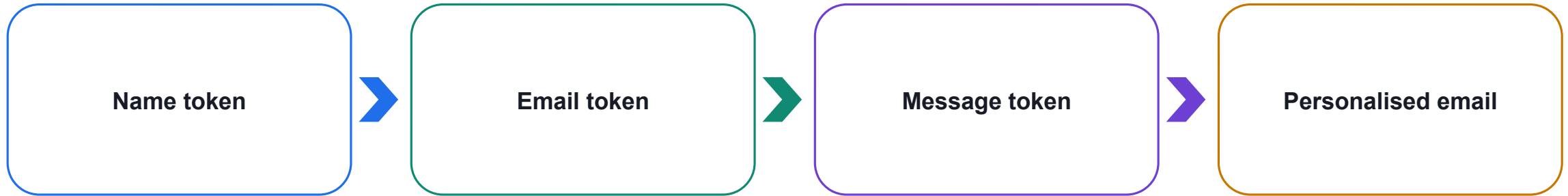
Tel provides an alternative contact channel.

3

Enquiry

Message uses a required long-answer field.

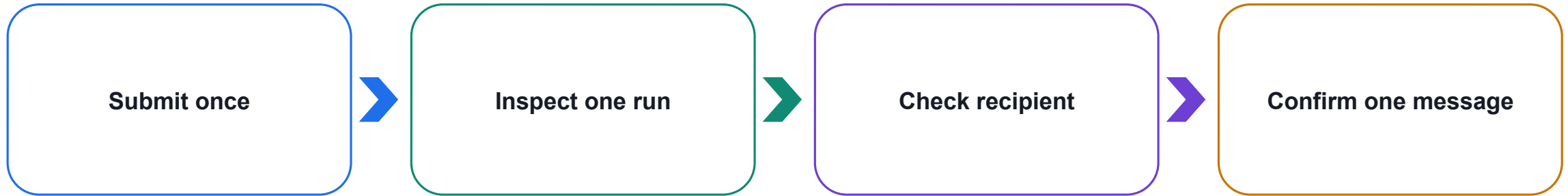
Lab 1 — Map Dynamic Content



Teaching point

Use the submitted Email answer as the recipient; do not type the flow owner's address.

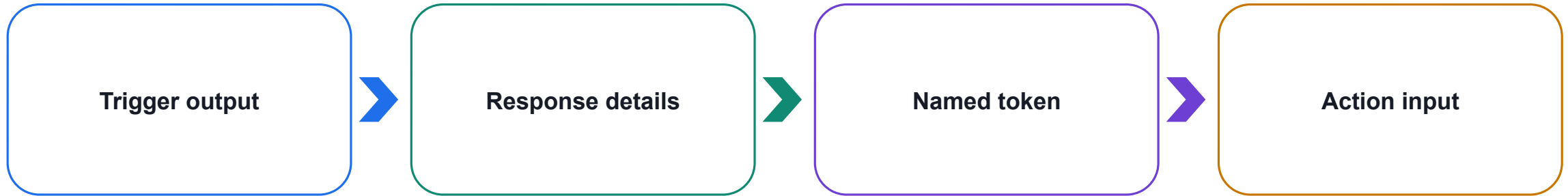
Lab 1 — Test the Real Outcome



Teaching point

One form submission must create one successful run and exactly one confirmation email.

Dynamic Content Carries Data Between Actions



Teaching point

Typed labels are static text; dynamic-content tokens contain values from the current run.

Excel Online Requires Structured Data

1

Workbook

Store the file in OneDrive for Business or SharePoint.

2

Table

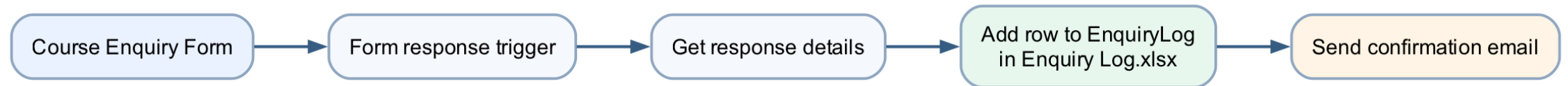
Use a named table with stable column headers.

3

Audit row

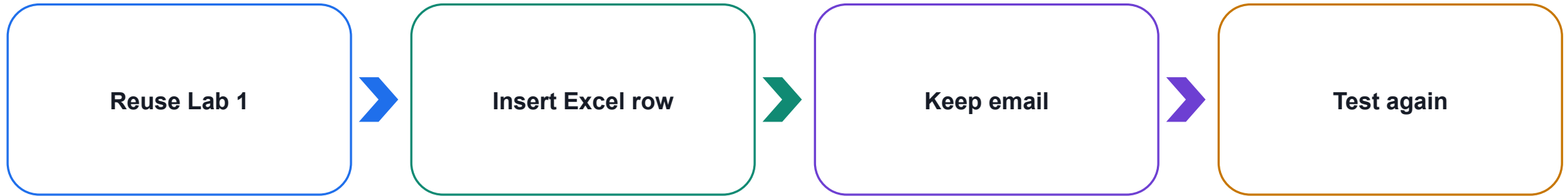
Record timestamp, source, status and the submitted answers.

Lab 2 — Log the Enquiry and Send Email



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

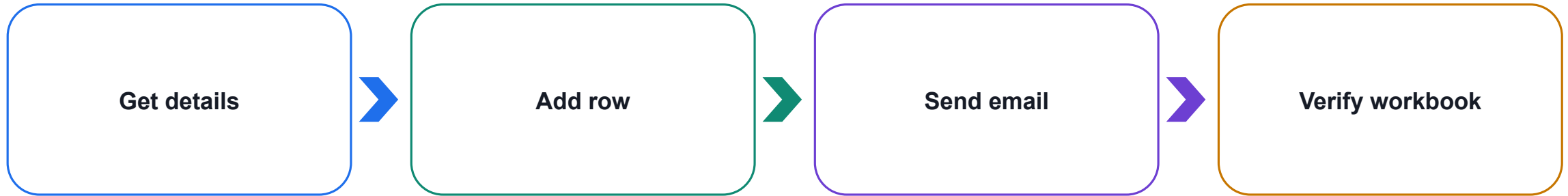
Lab 2 — Extend the Working Flow



Teaching point

Preserve the working trigger and email, then insert the durable audit record.

Lab 2 — Order Actions for Reliability



Teaching point

Send confirmation only after the workbook row succeeds, so the message matches reality.

Lab 2 — Troubleshooting

1

Location

Use OneDrive for Business or SharePoint, not a local path.

2

Table

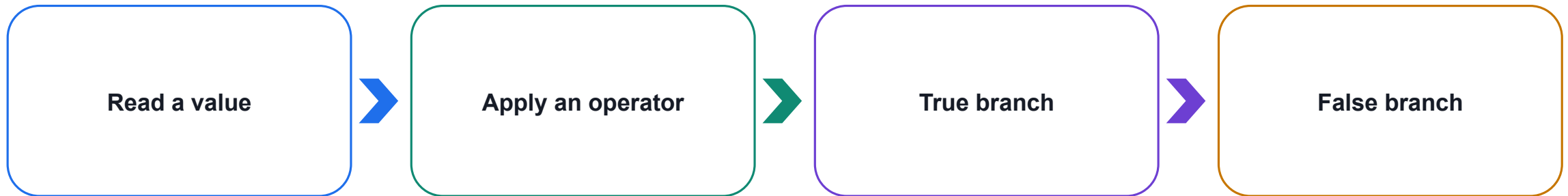
Select EnquiryLog; ordinary worksheet cells are not enough.

3

Lock

Close desktop Excel if the connector reports a file lock.

Conditions Turn Data into Decisions



Teaching point

A condition should express one business decision clearly and route mutually exclusive outcomes.

Design and Test Both Branches

1

Input

Use a controlled choice such as Yes/No to reduce ambiguity.

2

Branches

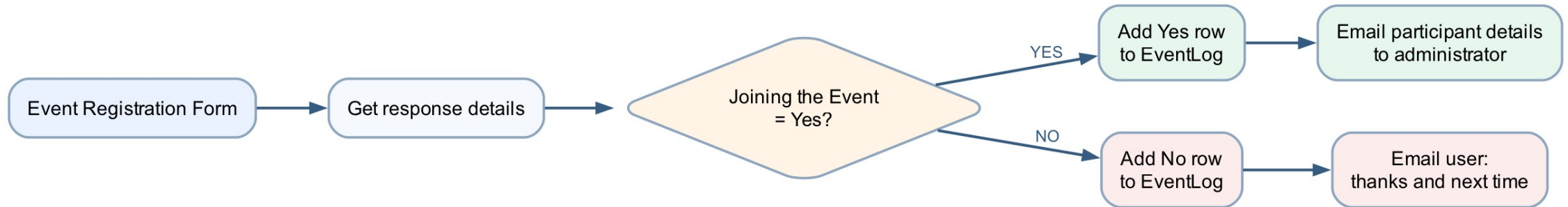
Give each path a complete action sequence and expected result.

3

Evidence

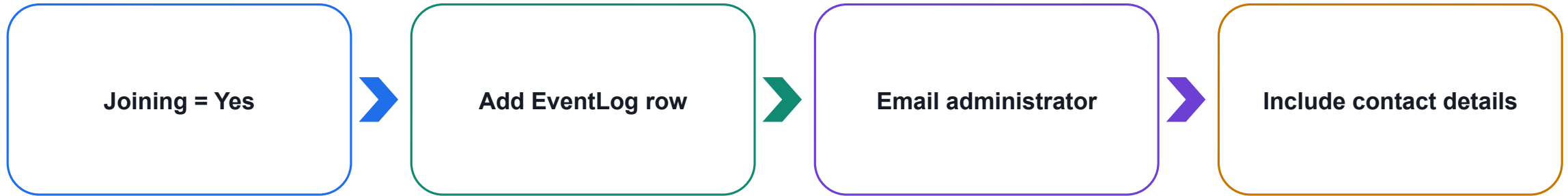
Run one test per branch and compare the actual outputs.

Lab 3 — Event Registration Branching



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

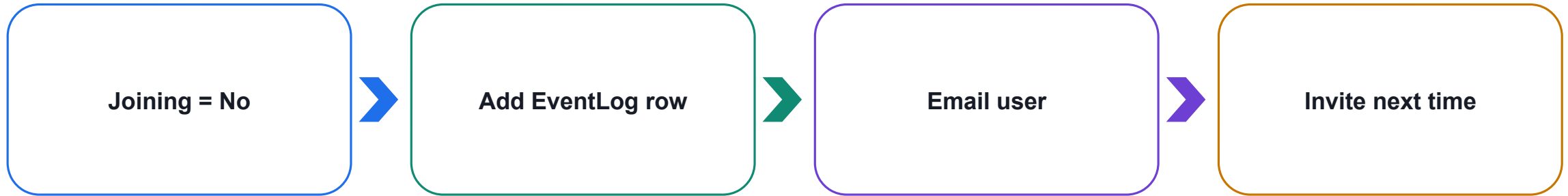
Lab 3 — Yes Branch



Teaching point

The administrator receives the participant's name, email and telephone number.

Lab 3 — No Branch



Teaching point

The user receives thanks and a respectful invitation to a future event.

Lab 3 — Branch Evidence

1

Yes test

Correct EventLog row and administrator email.

2

No test

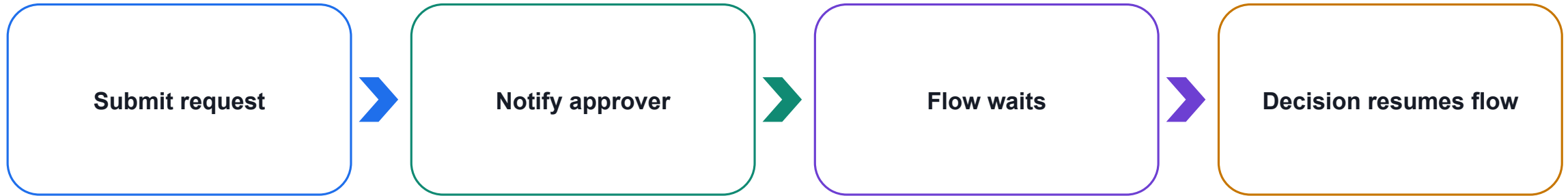
Correct EventLog row and user email.

3

Audit

JoiningEvent and NotificationSent explain the outcome.

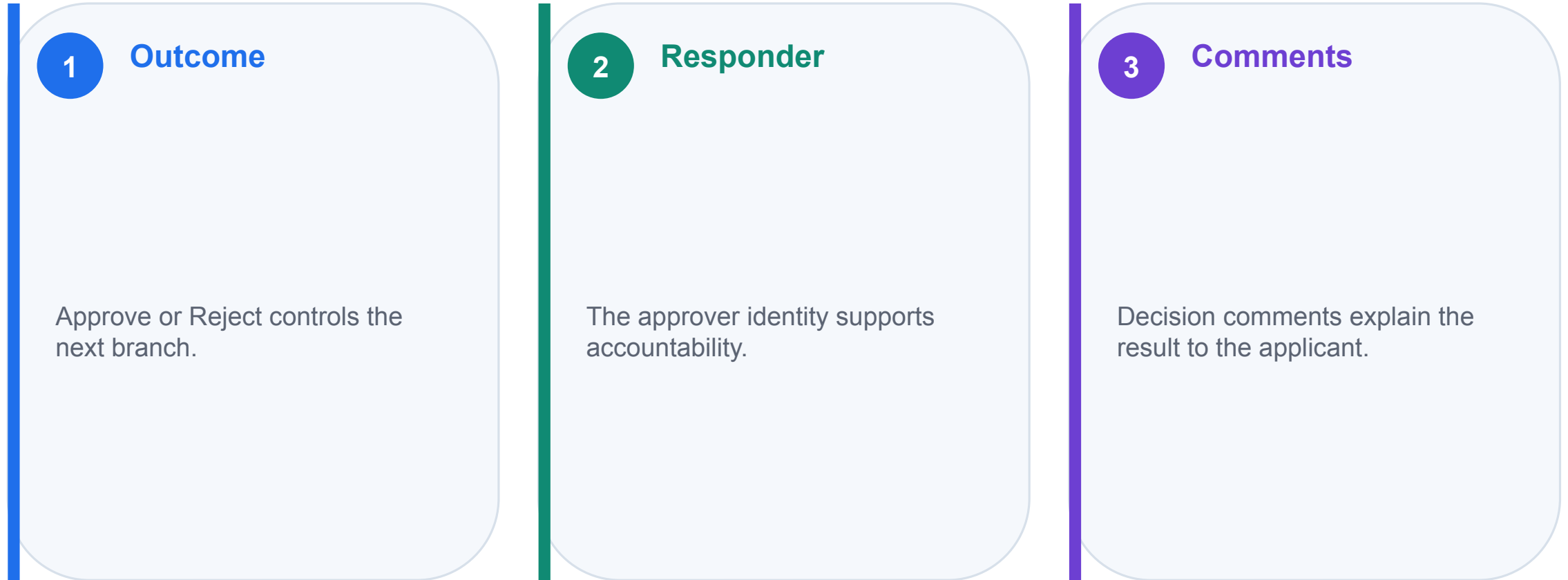
Approvals Add Human Judgement to Automation



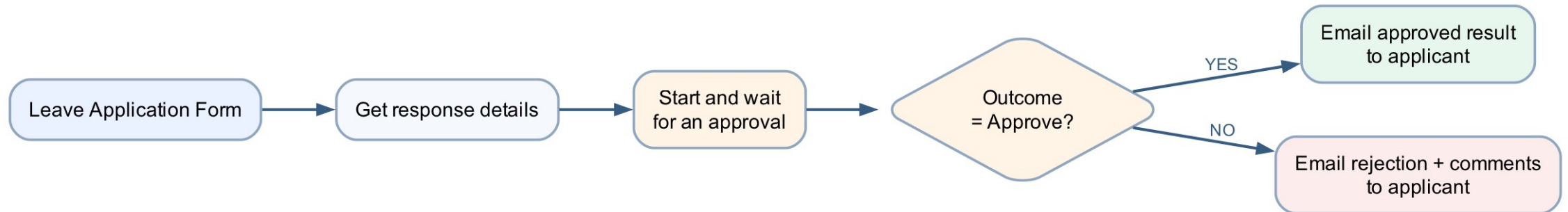
Teaching point

Use an approval when a named authorised person—not an AI model—must own the decision.

An Approval Produces Structured Evidence



Lab 4 — Leave Application Approval



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 4 — Leave Form

1

Employee

Name; Forms captures the responder identity.

2

Dates

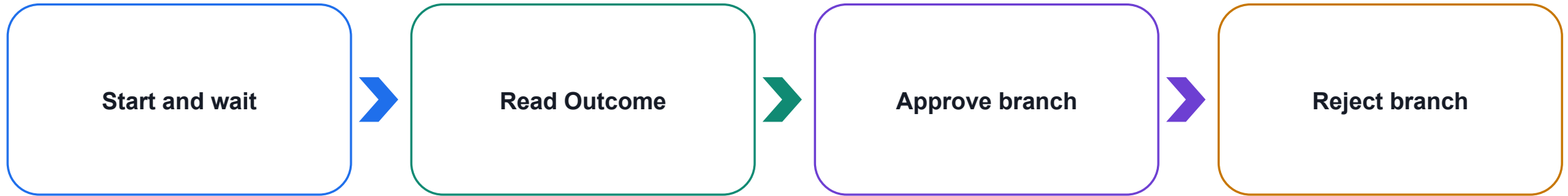
Leave start and end date.

3

Request

Leave Type and Reason for Leave.

Lab 4 — Handle the Approval Outcome



Teaching point

Compare Outcome to Approve and include the manager's comments in the applicant email.

Lab 4 — Governance

1

Access

Use only authorised approvers and accounts.

2

Privacy

Limit medical or unnecessary personal information.

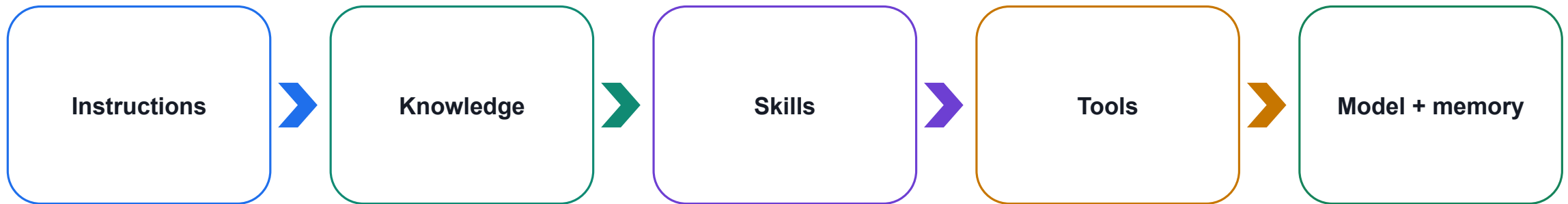
3

Evidence

Retain the decision, comments and flow run history.

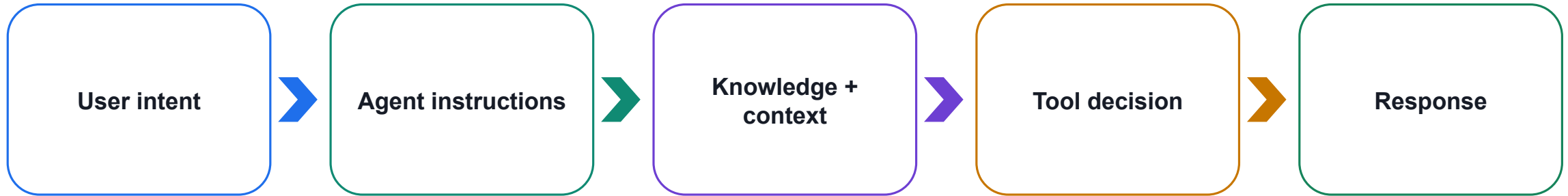
Copilot Studio Agents — Understand, Ground and Act

AGENTS



Learn the six building blocks before creating the IT and HR support agents.

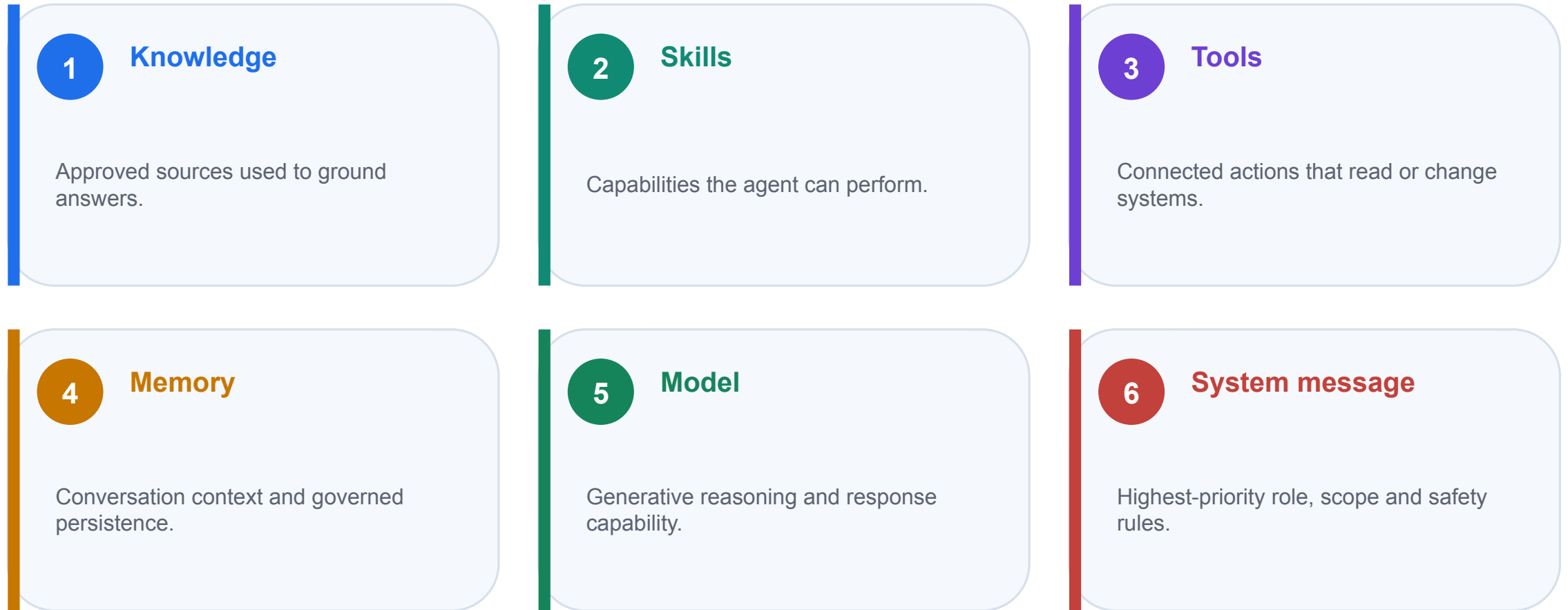
What Is a Copilot Agent?



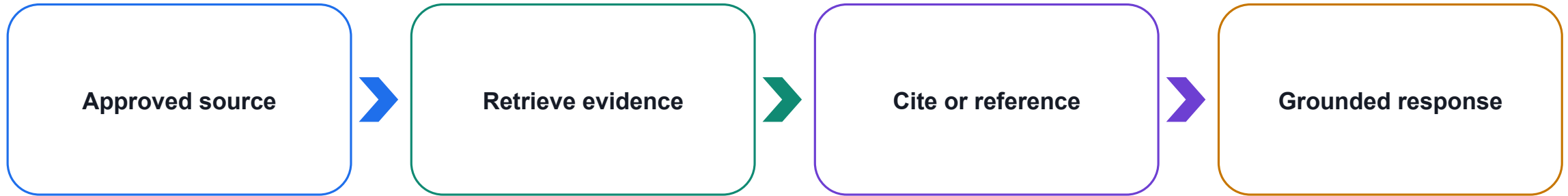
Teaching point

An agent interprets language and coordinates approved knowledge and actions inside defined boundaries.

Six Building Blocks of a Copilot Agent



Knowledge Grounds the Answer



Teaching point

Knowledge answers what the agent should know; source quality and permissions determine trust.

Skills Describe What the Agent Can Do

1

Conversation

Ask, clarify, classify, summarise and explain.

2

Reasoning

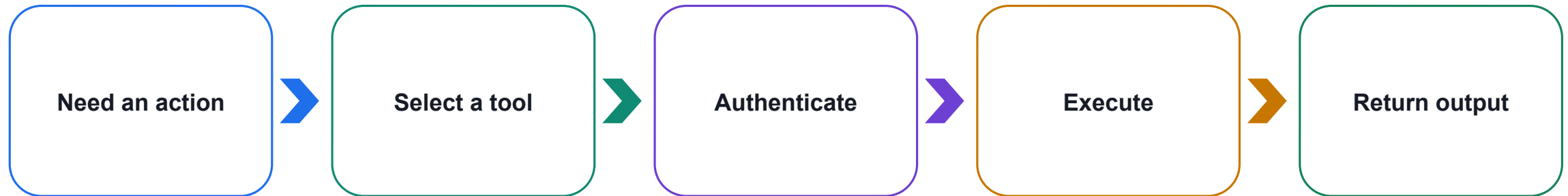
Apply instructions and knowledge to the current request.

3

Execution

Use topics, tools and agent flows to complete work.

Tools Let the Agent Act



Teaching point

A tool reads live data or creates an outcome; grant only the minimum authorised capability.

Memory Is Governed Context

1

Session context

Recent messages help the agent understand follow-up questions.

2

Variables

Named values preserve structured facts during the conversation.

3

Persistence

Longer-term memory depends on enabled features, consent and policy.

The Model Provides Generative Capability

1

Quality

Can it interpret the request and produce the required format?

2

Latency

Can it respond within the experience and workflow limits?

3

Governance

Does its use satisfy organisational and data requirements?

The System Message Defines Non-Negotiable Behaviour



Teaching point

Copilot Studio may label this control Instructions; treat it as policy, not decoration.

Test Before Publishing an Agent

1

Expected

Questions the agent should answer from approved sources.

2

Unsupported

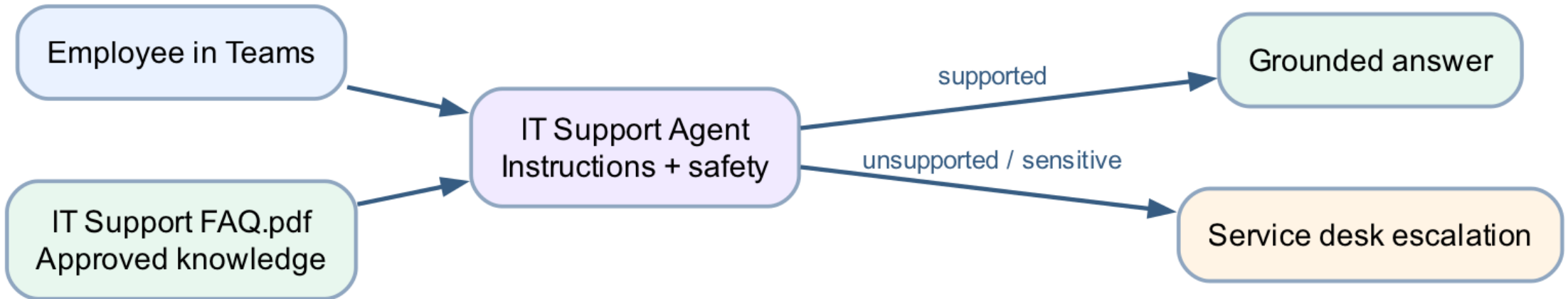
Missing or out-of-scope information that requires a limitation.

3

Unsafe

Credentials, private data or prohibited decisions that must be refused.

Lab 5 — IT Support Agent



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 5 — Ground the IT Support Agent

1

Role

First-line support for password, MFA, VPN, Wi-Fi and device loss.

2

Knowledge

Upload only the approved IT Support FAQ.

3

Boundary

Never request passwords, MFA codes or recovery keys.

Lab 5 — Test and Deploy to Teams

1

Positive

Verify grounded answers for approved IT questions.

2

Negative

Refuse credentials and unrelated HR requests.

3

Teams

Publish the latest version and verify the installed agent.

SharePoint Knowledge Adds Governance

1

Managed source

Policy owners update one controlled document library.

2

Permissions

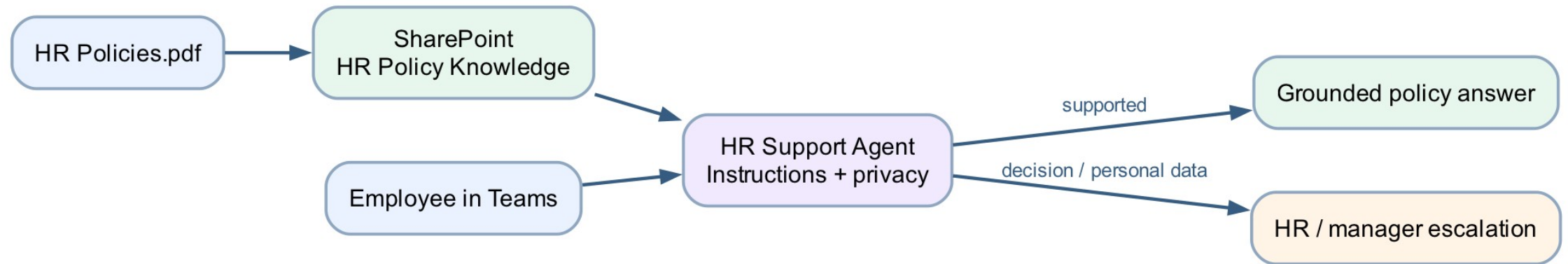
The agent can use only content its connection may access.

3

Freshness

Re-test the agent whenever source policies change.

Lab 6 — HR Support Agent



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 6 — Ground the HR Support Agent

1

Explain

Use approved leave, expenses and workplace-policy content.

2

Protect

Never expose another employee's records or infer private facts.

3

Escalate

HR and managers retain final authority for personal decisions.

Lab 6 — Publish and Verify in Teams

1

Source test

Confirm the response is traceable to the SharePoint policy.

2

Boundary test

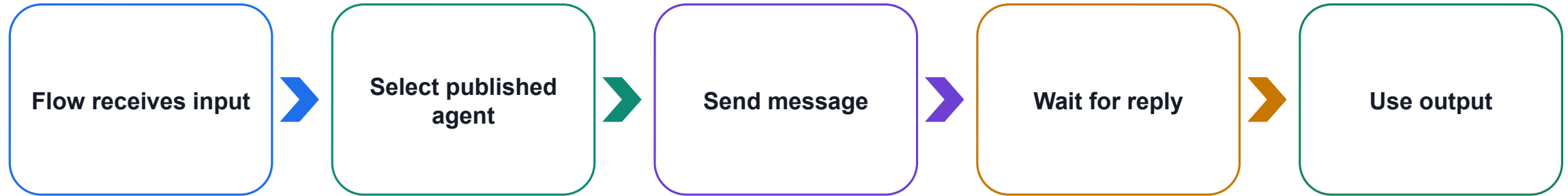
Ask for another employee's record and verify refusal.

3

Channel test

Publish the current version and verify Teams behaviour.

Power Automate Can Execute a Published Agent



Teaching point

The flow remains the process controller while the specialised agent handles grounded interpretation.

Route to the Specialist with the Right Knowledge

1

Classify

Use the submitted Support Type choice.

2

Invoke

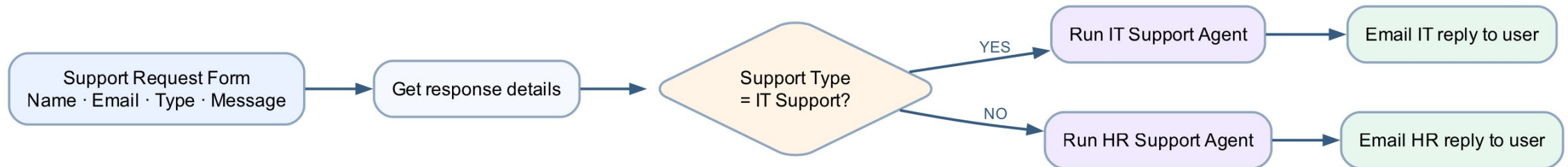
Call only the IT or HR agent selected by the condition.

3

Return

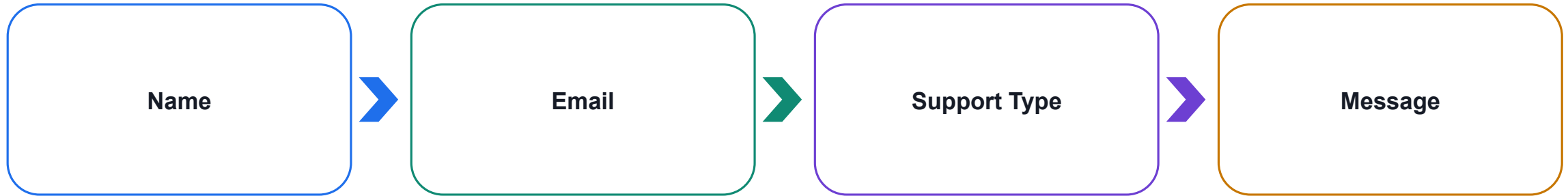
Email only the reply produced by the chosen branch.

Lab 7 — Support Request Routing



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

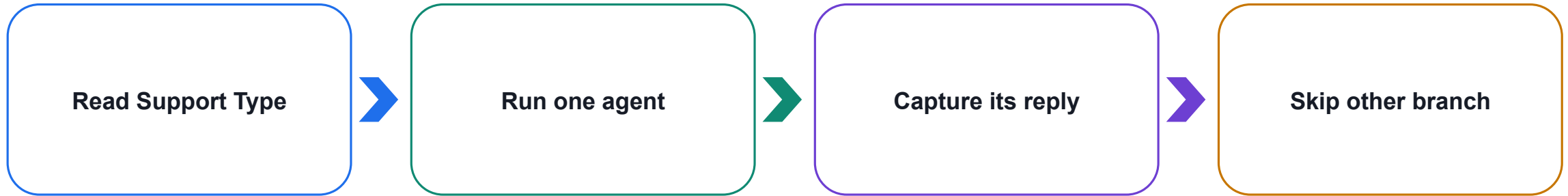
Lab 7 — Build the Support Request Form



Teaching point

A controlled IT Support / HR Support choice makes the routing rule deterministic.

Lab 7 — Execute Only the Selected Agent



Teaching point

One submission should create one agent execution and one user email—not two.

Lab 7 — Test the Complete Routing Matrix

1

IT test

IT question → IT agent → IT-grounded email.

2

HR test

HR question → HR agent → HR-grounded email.

3

Fallback

Invalid or unsafe requests receive a bounded response.

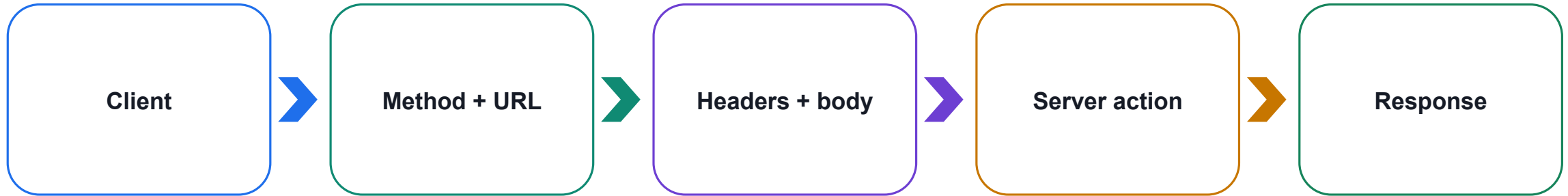
Day 2 — HTTP, Webhooks and Agent Websites

DAY 2



Learn the web integration contract before connecting websites and finance agents.

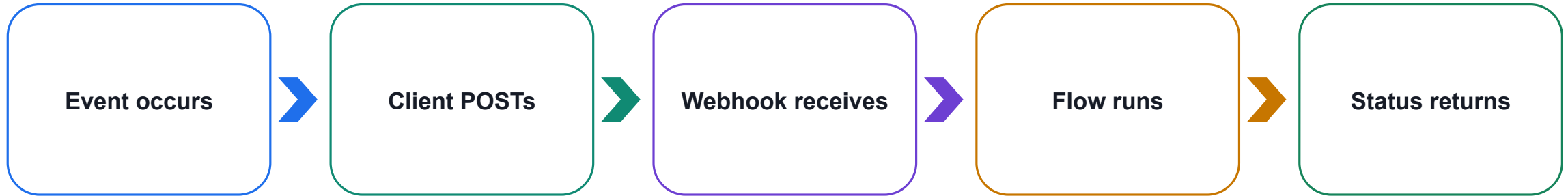
An HTTP Request Is a Message Between Systems



Teaching point

The course websites act as clients; Power Automate receives the request and returns a bounded response.

A Webhook Is an Event-Receiving HTTP Endpoint



Teaching point

Power Automate generates the webhook URL only after a valid trigger-and-action flow saves.

Common HTTP Methods

1

GET

Read data

2

POST

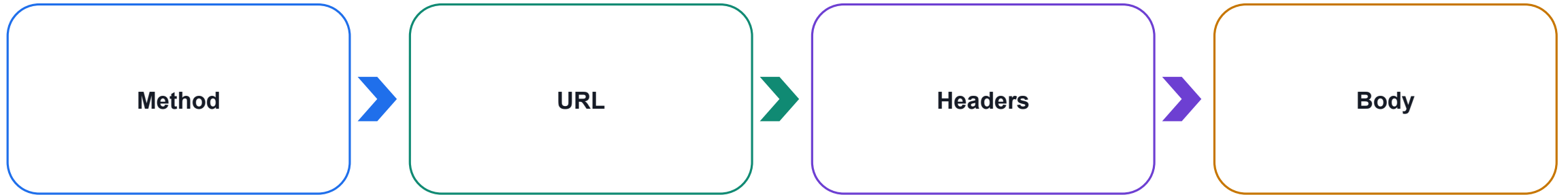
Submit data or start work

3

PUT · PATCH · DELETE

Update or remove data

An HTTP Request Has Four Main Parts



Teaching point

The lab websites use POST to send JSON data to the learner's Power Automate endpoint.

JSON Defines a Machine-Readable Contract

1

Schema

Field names, data types and required values define valid input.

2

Payload

The browser serialises the current form or prompt values.

3

Response

The flow returns limited JSON the page can display.

Webhook Security and Browser Boundaries

1

Endpoint

Treat an anonymous webhook URL like a secret.

2

Input

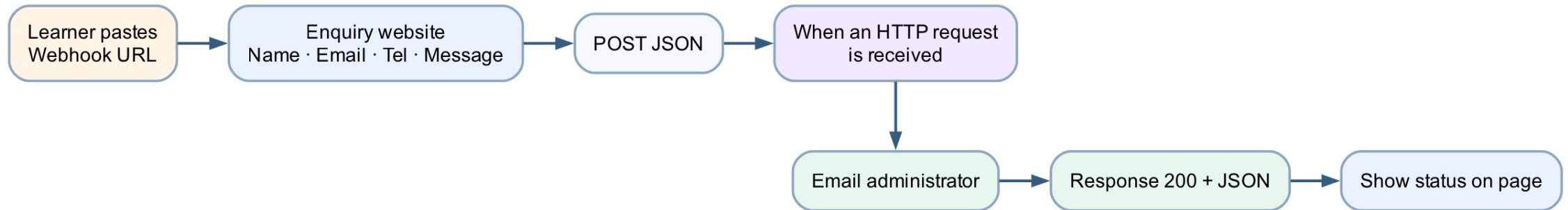
Validate, minimise and never include credentials.

3

CORS + output

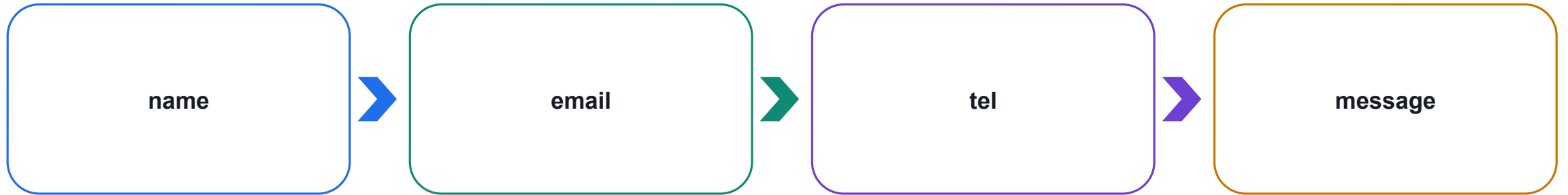
Return only what the browser needs; authenticate production callers.

Lab 8 — Website HTTP Enquiry



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 8 — Define the Request Contract



Teaching point

Generate the trigger schema from a representative sample before mapping fields.

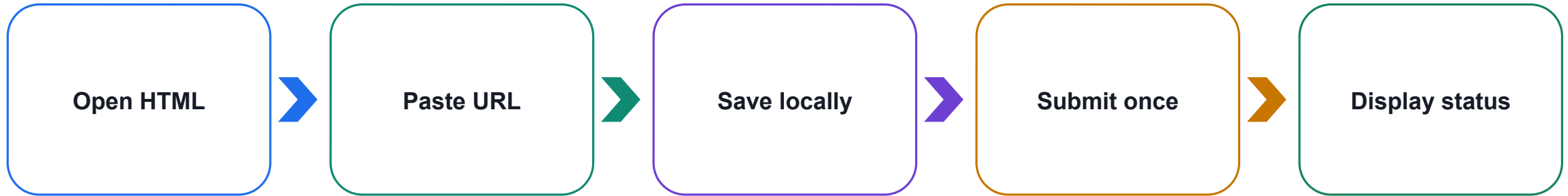
Lab 8 — Complete the Flow Before Saving



Teaching point

A trigger alone is incomplete; add actions and resolve validation errors before expecting a URL.

Lab 8 — Connect the Supplied Website



Teaching point

The supplied source contains no tenant endpoint; each learner connects their own saved flow.

Lab 8 — End-to-End Evidence

1

Browser

The page reports one successful response.

2

Flow

Run history shows the expected request fields.

3

Mailbox

The administrator receives exactly one enquiry email.

Grounded Finance Answers Need Retrieval

1

Retrieve

Search the approved Finance Knowledge Base for relevant passages.

2

Generate

Compose an explanation constrained by the retrieved evidence.

3

Qualify

State uncertainty and refuse guarantees or personalised advice.

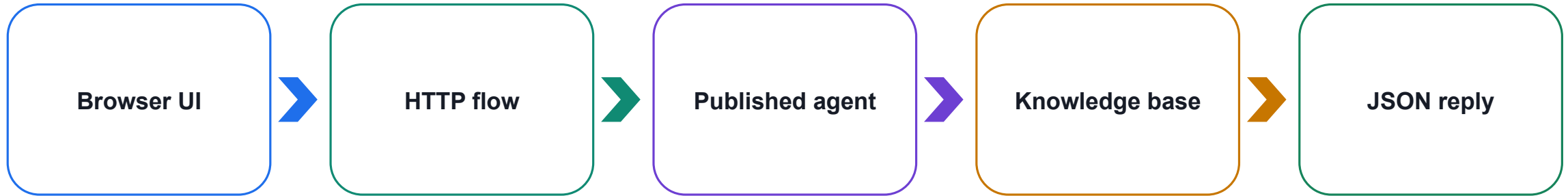
Execute Agent and Wait Bridges Flow and Agent



Teaching point

Use the actual agent-response token; do not copy a test answer into the flow.

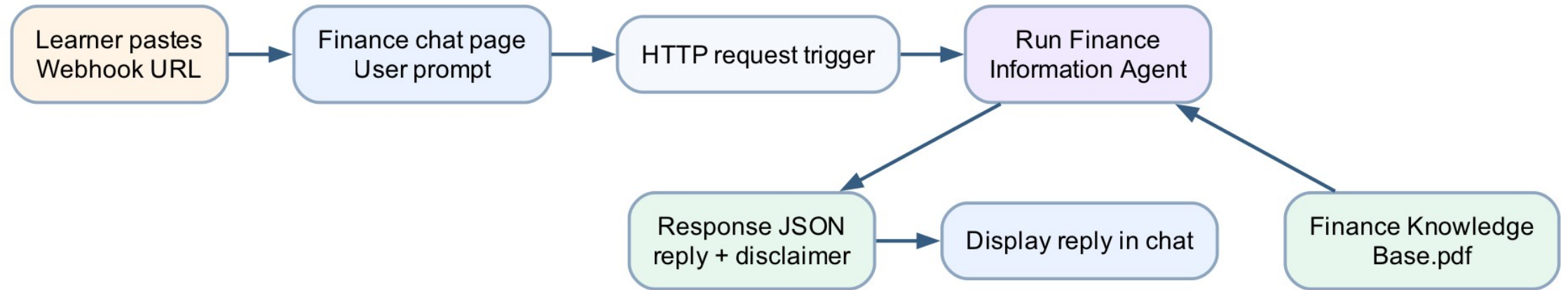
Web Chat Is a Channel, Not the Knowledge Source



Teaching point

The page captures and displays; the flow orchestrates; the agent interprets approved knowledge.

Lab 9 — Finance Agent Web Chat



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 9 — Finance Knowledge Boundary

1

Explain

Orders, candles, timeframes, volatility and news concepts.

2

Qualify

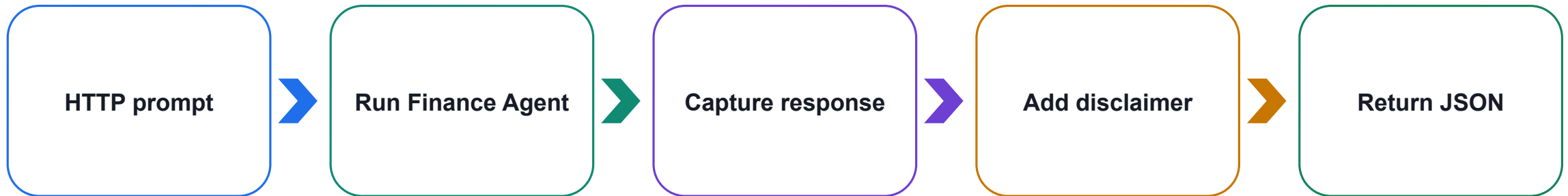
Separate facts from interpretation and disclose uncertainty.

3

Refuse

No guaranteed returns or personalised buy/sell instructions.

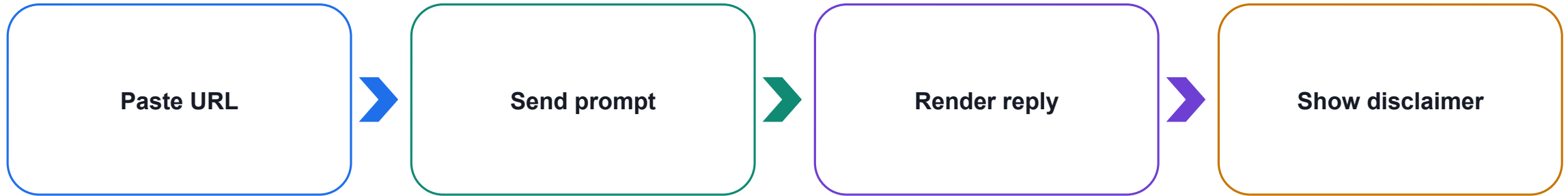
Lab 9 — Build the HTTP-to-Agent Flow



Teaching point

One browser message should create one flow run and one bounded agent response.

Lab 9 — Connect the Finance Chat Page



Teaching point

Store the URL only in the learner's browser and never commit it to the repository.

Lab 9 — Grounding and Safety Tests

1

Known concept

Ask about a market versus limit order.

2

Uncertainty

Ask something absent from the knowledge source.

3

Boundary

Request a guaranteed return and verify refusal.

Tools and APIs Supply Live Context

1

Tool

A governed function the agent or flow is authorised to call.

2

API

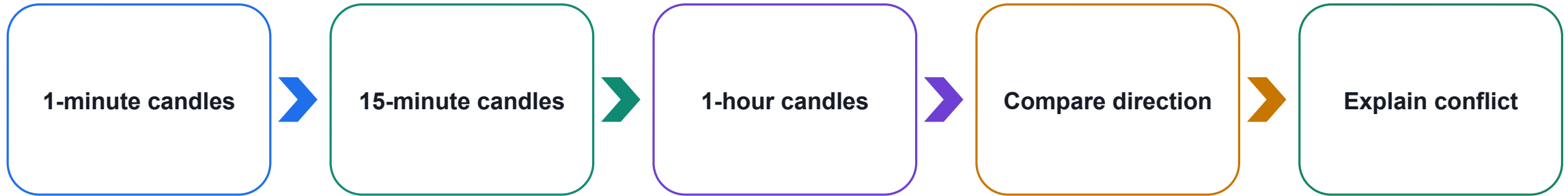
A structured contract for requesting data from an external service.

3

Output

Validated tool data becomes evidence—not an automatic conclusion.

Multi-Timeframe Analysis Compares Different Horizons



Teaching point

Short intervals contain more noise; a responsible answer preserves conflicting evidence.

News Adds Context, Not Certainty

1

Time

Check when the event and article were published.

2

Source

Prefer relevant, credible reporting and identify gaps.

3

Impact

Relate news cautiously; price may already reflect it.

Keep Credentials on the Server Side

1

Store

Use protected connections, environment variables or a custom connector.

2

Protect

Enable secure inputs and outputs where credentials appear.

3

Never expose

No API keys in HTML, prompts, source control or screenshots.

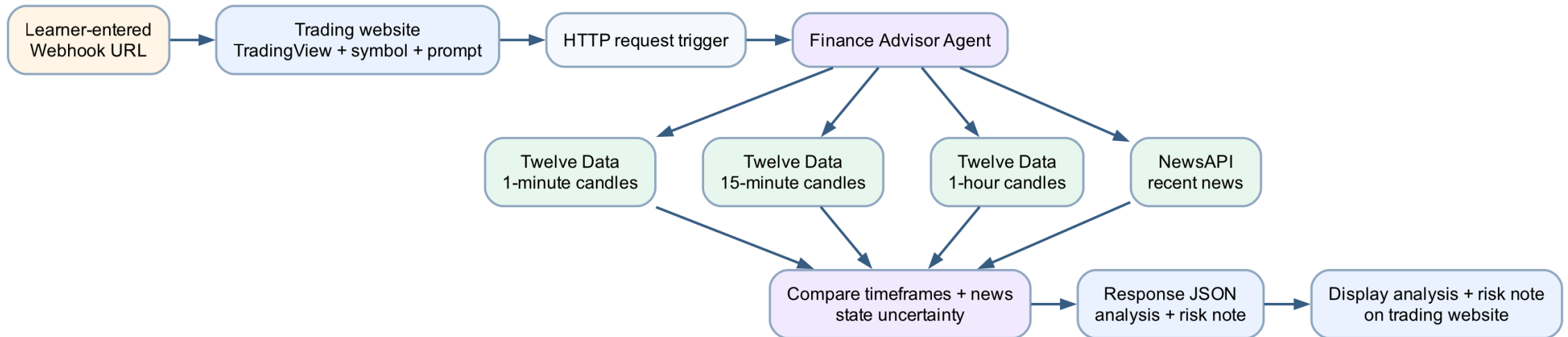
Guardrails Constrain the Finance Advisor



Teaching point

The agent provides educational analysis and never promises returns or gives personalised instructions.

Lab 10 — AI Trading Advisor Website



Shared workflow visual — follow the arrows from the trigger through actions to the expected outcome.

Lab 10 — Authorised Data and Agent Tools

1

Twelve Data

Retrieve 1-minute, 15-minute and 1-hour candles.

2

NewsAPI

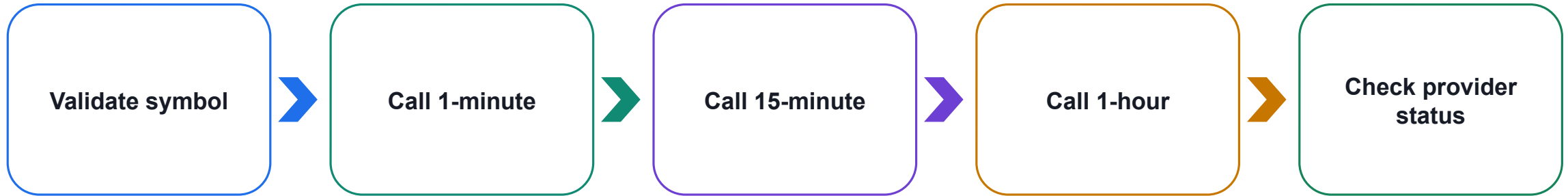
Retrieve recent symbol or company news.

3

Finance Advisor

Compare evidence and return bounded analysis.

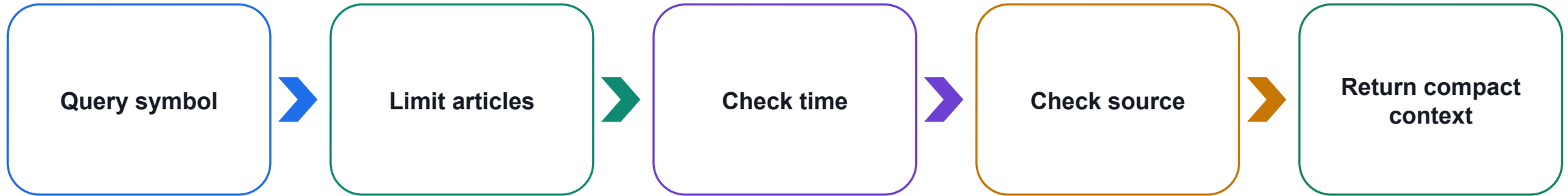
Lab 10 — Build the Three Candle Requests



Teaching point

Use the same symbol and comparable result limits so the agent can contrast the intervals.

Lab 10 — Retrieve Recent News



Teaching point

Keep the news payload small and preserve provider errors instead of hiding them.

Lab 10 — Secure the External Calls

1

Connection

Keep provider credentials in protected server-side configuration.

2

Observability

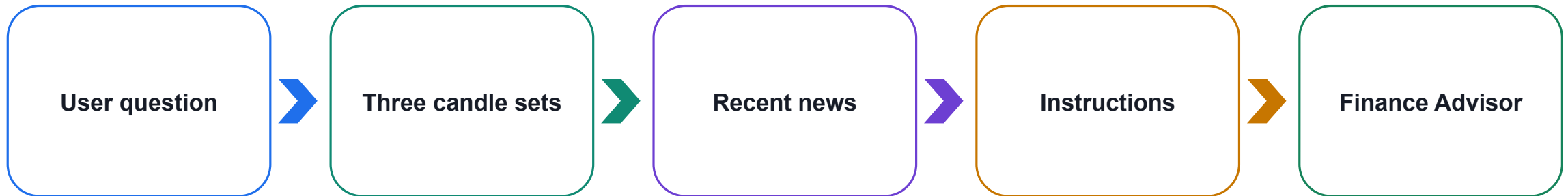
Log status and latency without recording secrets.

3

Failure

Return a safe partial-result message when a provider is unavailable.

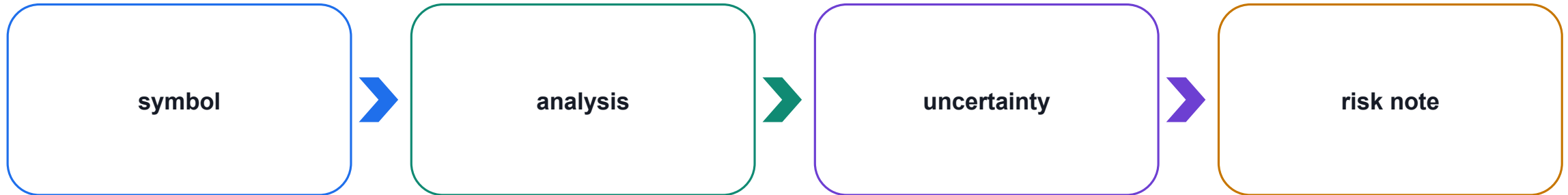
Lab 10 — Compose the Agent Input



Teaching point

Tell the agent which observations are data, which are user context, and which rules are mandatory.

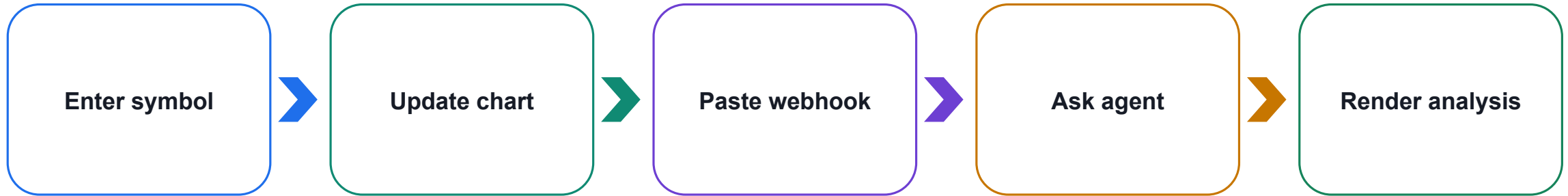
Lab 10 — Return Structured JSON



Teaching point

A stable response contract lets the website render success and errors predictably.

Lab 10 — Embed TradingView Separately



Teaching point

TradingView visualisation and agent analysis are related but independently testable components.

Lab 10 — Test the Complete Matrix

1

Valid

All tools succeed and the response is qualified.

2

Partial

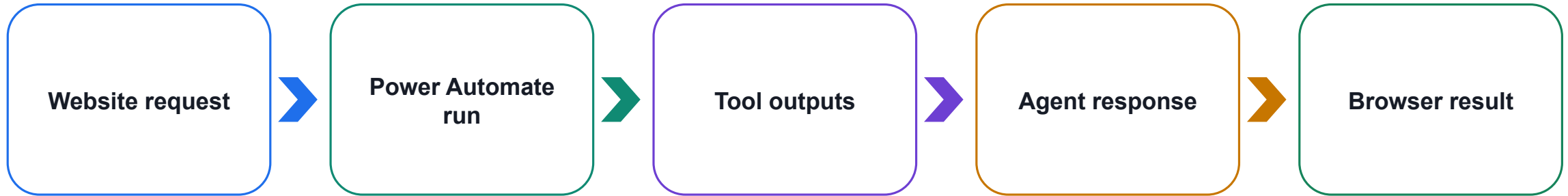
Missing news or conflicting timeframes are disclosed.

3

Invalid

Bad symbol or advice request receives a safe response.

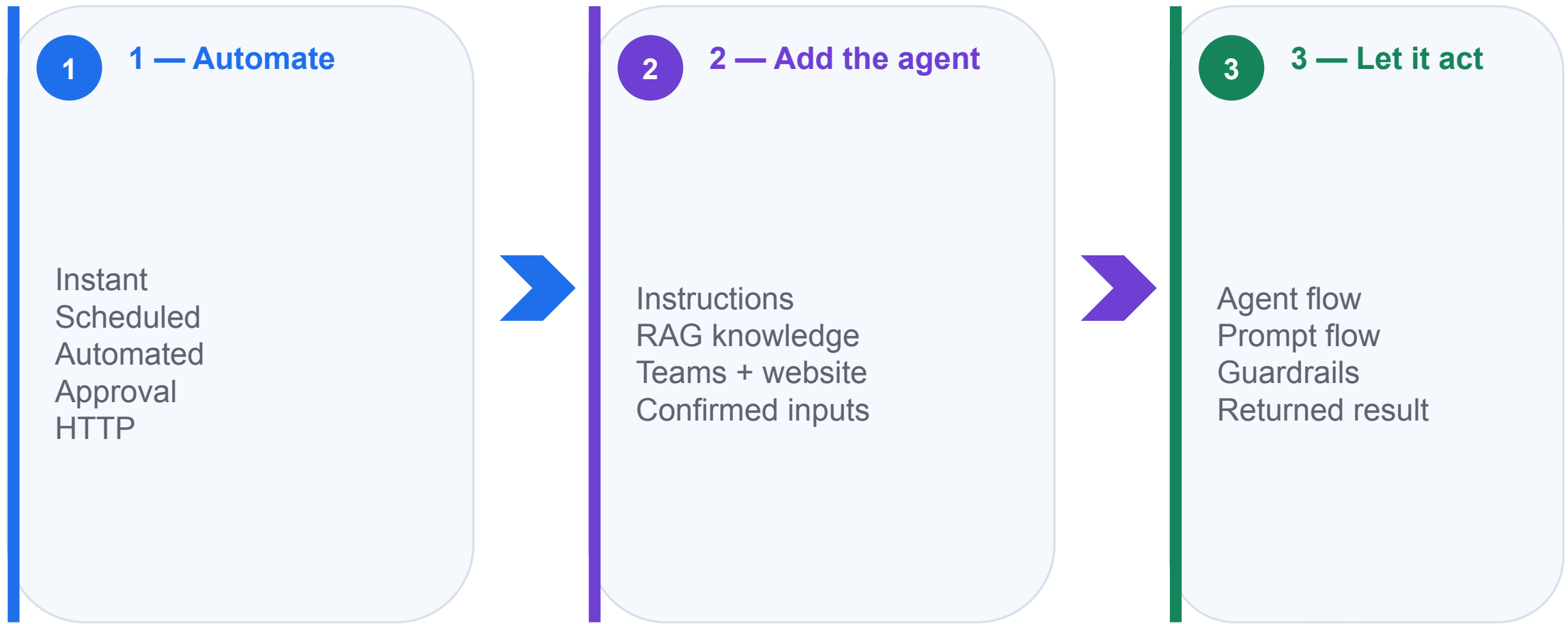
Lab 10 — Review the End-to-End Trace



Teaching point

Each boundary must have a clear owner, observable input/output and safe failure behaviour.

Course Recap — The Big Picture



What You Can Automate Next

1

Onboarding: new-hire checklists and welcome emails

2

Reporting: scheduled data pulls and summaries

3

Helpdesk: route requests to the right specialist agent

4

Document handling: filing, approvals and reminders

5

Agent-assisted workflows: interpret language, then perform governed actions

Governance and Operational Readiness

1

Use authenticated endpoints, least privilege and protected credentials

2

Validate inputs and return only the minimum necessary output

3

Keep approvals and high-impact decisions with authorised people

4

Monitor failures, latency, connector health and source freshness

5

Document ownership, evidence, escalation and retirement procedures

Key Takeaways

1

Trigger choice determines instant, scheduled or automated flow behaviour

2

Actions consume dynamic data and produce observable business outcomes

3

Conditions and approvals encode rules while preserving human authority

4

Copilot agents combine instructions, knowledge, skills, tools, memory and a model

5

HTTP and webhooks connect external experiences to governed flows and agents

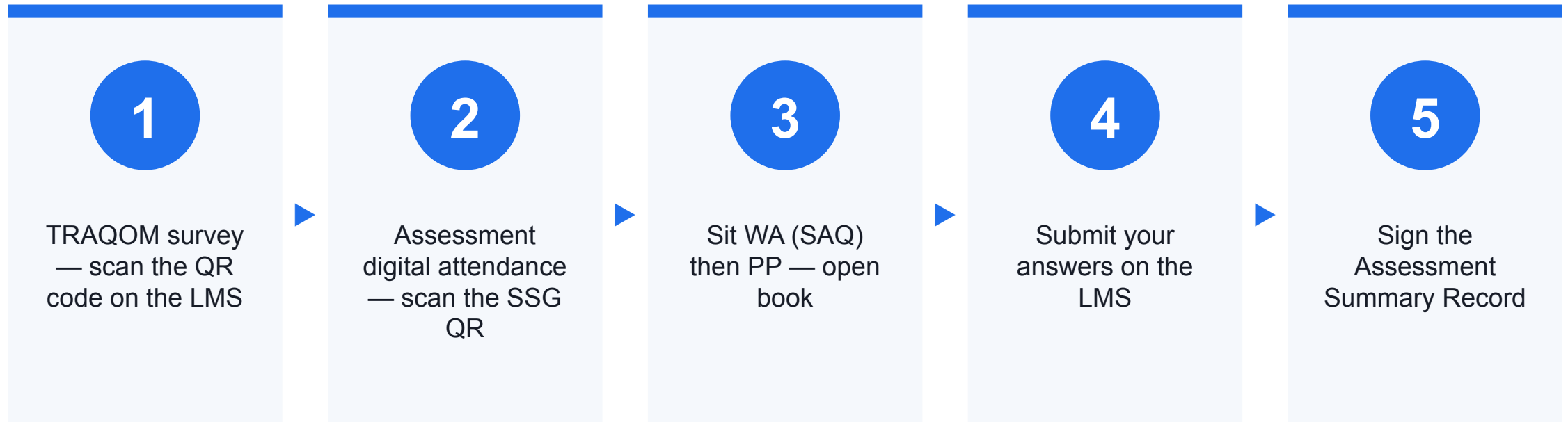
Courseware & Assessment on the LMS

- Download the slides and the Learner Guide from the LMS for the open-book assessment.
- Portal: <https://lms-tms.tertiaryinfotech.com/>
- Submit your Written Assessment and Practical Performance answers on the LMS.
- Complete the TRAQOM survey via the QR code on the LMS.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Thank You

Now go automate something — questions welcome